

Introduction to computing

Nested list:-

(2-dimensional list) but proper name is nested list.

```
pairs = [
```

```
    [1, 20],
```

```
    [2, 30],
```

```
    [3, 40],
```

```
    [4, 50],
```

```
]
```

outer list
index

Pair [2][1]

, pair [0][0]

↓

↓

40

1

out
put


```
pairs = [[1,2], [2,2], [2,3], [4,4]]
```

```
same_count = 0
```

```
for x, y in pairs:
```

```
    if x == y
```

```
        same_count += 1
```

```
print same_count
```

Operations on list:-

```
a = [66.25, 333, 333, 1, 1234.5]
```

```
    ↑  
    print a
```

```
value new ← a.append(987)
```

```
    print a
```

```
    ↓  
    [66.25, 333, 333, 1234.5, 987]
```


a = [1, 2, 3, 8, 9]

↑
print a

a.insert(2, -1)

↓ 2 wala index ko agay push kary ga
or (-1) waha pa insert karay ga.

print a
↓

[1, 2, -1, 3, 8, 9]

print a

a.index(9)

↓ position batay ga 9 ki

↓
5

} print a
a.remove(3)
print a

print a

a.reverse()

print a → reverse the array

print a

a.sort()

print a

a = [1, 2, 3, 4, 10, 9]

print a

a.sort(reverse=True)

print a

↓

[10, 9, 4, 3, 2, 1]

help(list)

↓

for all the functions
in list-

Strings are also sequence

s = "I am a dummy string. lorem ipsum"

a[2]

a

len(s)

33

s[1] = 'u'

→ ham string ma aisaay
element insert nahi kar saktay

help(str)

Ranges:-

`range(1, 10)` ^{by default}
one.

`[1, 2, 3, 4, 5..., 9]`

`range(1, 10, 2)`
↳ step, size

`[1, 3, 5, 7, 9]`

} `for i in range(1, 3)`
`print 2, "x", i,`
`"=", 2 * i`

List comprehensions :-

`nums = [1, 2, 3, 4, 5, 6]`

`evens = []`

`for i in nums:`

`if i % 2 == 0`

`evens.append(i)`

`print evens`

} `evens = [i for i in nums`

`if i % 2 == 0]`

`print evens`

Tuples - immutable

data types :-

digits = (1, 8, 2, 8)

always define
by brackets

len(digits)

digits [1]

8

digits[0] = 11 -> we cannot assign item to
tuple.